Lecture plan · for the lecturer

Transactions and distributed data

Data engineering · University · Lecture 3

Time	Materials
60 min	Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- State the ACID guarantees.
- Recognise isolation anomalies.
- Argue CAP trade-offs.

Activity: Interleave and fix

1. Warm-up (5 min): Show both confirmation emails and ask pairs to explain the bug.
2. Act (15 min): Two volunteers play T1 and T2 while the class calls out interleavings.
3. Reproduce (20 min): Pairs reproduce the lost update on a real database, then fix it.
4. Argue (15 min): Decide what registration and the library catalogue should do during a partition.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which interleavings were safe?
- What is your default isolation level?
- Which system chose availability?

Exit ticket

- Name one operation that must stay consistent during a partition.

Try it in your next lecture

Give two volunteers cards with the steps of two transactions; the class calls out an interleaving and records the final seat count, then add a lock and try again.

Seminar sheet · for students

Transactions and distributed data

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Name _____ Class _____ Date _____

1 **During a partition, a ledger must never show two balances. What does it give up?**

◯ Consistency ◯ Availability ◯ Durability

2 **Count the outcome.**

seats = 1. T1 and T2 both read 1, both write seats - 1 and enrol a student. Final seats = _____ and students enrolled = _____.

Word bank: 0 · 1 · -1 · 2

3 **Which prevent the double booking? Choose all that apply.**

☐ UPDATE … SET seats = seats - 1 WHERE seats > 0, checking rows affected

☐ Serializable isolation with retry ☐ SELECT … FOR UPDATE before writing

☐ Reading the count twice

4 **Snapshot isolation guarantees serializability.**

◯ True ◯ False

5 **Match each anomaly to the weakest level that prevents it in the ANSI model.**

Dirty read	• • • •	Serializable
Non-repeatable read	• • • •	Repeatable read
Phantom	• • • •	Read committed

✎ **Which isolation level does the database your students use default to, and have they ever seen it fail?**